

# Malnutrition Detection Using Facial Images

## Deep Learning Model Progress Report

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## 1. Introduction

Malnutrition detection in children is a critical healthcare challenge, particularly in regions where access to clinical screening and diagnostic infrastructure is limited. Traditional approaches rely on anthropometric measurements such as height, weight, and body mass index, along with expert clinical evaluation. While effective, these methods are often time-consuming, require trained personnel, and may not be scalable in low-resource settings.

Recent advancements in computer vision and deep learning have opened the possibility of using facial images as a non-invasive and scalable alternative for health assessment. Facial features can capture subtle structural and physiological cues associated with malnutrition, making them a viable input for automated classification systems.

This project is based on the implementation and exploration of a research study that utilises deep convolutional neural networks for malnutrition detection using facial imagery. The goal is not only to reproduce the methodology proposed in the paper but also to extend it through experimentation with different architectures, training strategies, and hybrid approaches.

The primary objective of this work is to develop models capable of accurately distinguishing between normal and malnourished individuals while analysing how performance evolves across different architectural choices and training refinements. In this context, accuracy serves as a direct measure of classification correctness, indicating how often the model makes the right prediction. However, accuracy alone can be misleading in imbalanced datasets. Therefore, the Area Under the Receiver Operating Characteristic Curve (AUC) is also used as a key metric, as it reflects the model's ability to distinguish between classes across different decision thresholds. A higher AUC indicates stronger discriminative capability, making it particularly relevant for evaluating medical classification tasks where sensitivity and specificity are important.

## 2. Methodology

The dataset used in this project consists of approximately 446 facial images, with a noticeable class imbalance between normal and malnourished categories. Given the relatively small size of the dataset, training deep neural networks from scratch was not feasible. Instead, transfer learning was employed using pretrained convolutional neural networks, allowing the models to leverage feature representations learned from large-scale datasets.

To further enhance feature learning, an additional ageing dataset consisting of facial images of individuals across different age groups was incorporated for pretraining certain models. This step allowed the networks to learn more generalised facial representations before being fine-tuned on the malnutrition classification task. By doing so, the models were better equipped to capture subtle facial variations relevant to health conditions.

Three primary modelling approaches were explored. The first approach involved models based on ResNet-50, where progressive improvements were made through better training strategies and preprocessing. The second approach used DenseNet121 to evaluate how alternative architectures with different feature propagation mechanisms perform on the same task. The third approach involved hybrid models, where deep features extracted from pretrained networks were combined with classical machine learning classifiers such as Random Forests and Support Vector Machines.

All models were evaluated using accuracy and AUC, along with an analysis of classification behaviour through confusion matrices. This combination of evaluation metrics ensured that both overall performance and class-level discrimination were properly assessed.

### **3. Model-wise Progress**

#### **3.1 ResNet-50 Models**

The initial implementation was based on ResNet-50 using a standard transfer learning setup. The first model achieved an accuracy of approximately 34% with AUC close to 0.50, indicating near-random performance. The model showed strong bias, predicting almost all samples as a single class.

In the second model, training duration and optimisation were improved. This increased accuracy to approximately 41% and AUC to around 0.55. However, prediction bias persisted, showing that longer training alone was insufficient.

The third model introduced class balancing techniques and modified loss functions. Accuracy improved to approximately 50%, and AUC increased to around 0.60. While feature learning improved, predictions still collapsed toward a single class.

In the fourth model, dataset quality was improved by removing duplicate samples and ensuring consistent preprocessing. This resulted in more stable validation behaviour, with accuracy reaching approximately 60% and AUC around 0.65. Generalisation improved, but classification bias remained.

The fifth model applied selective fine-tuning of deeper layers along with weighted loss functions. Accuracy remained between 42–50%, but AUC improved further to approximately 0.65–0.68. This indicated better feature learning, although classification was still suboptimal.

A major breakthrough occurred in the sixth model when preprocessing was correctly aligned with the pretrained network. This led to a sharp increase in performance, with accuracy reaching approximately 83% and AUC around 0.80. The model demonstrated clear class separation and balanced predictions, making it the best-performing configuration.

Subsequent models explored ensemble-based extensions of ResNet. These achieved accuracies between 80–82% with AUC around 0.75–0.78. However, they required threshold tuning and showed variability across runs, indicating that increased complexity did not outperform the refined single-model approach.

### 3.2 DenseNet Models

To evaluate architectural differences, models based on DenseNet121 were implemented. The DenseNet model achieved an accuracy of approximately 76–77% with AUC around 0.76–0.77. It demonstrated stable learning and effective feature reuse, resulting in consistent class separation. However, it showed mild overfitting, with training performance exceeding validation performance. Overall, DenseNet provided strong and stable performance but did not surpass the best ResNet-based model.

### 3.3 Hybrid Models

A hybrid approach was implemented using feature extraction from multiple architectures, including DenseNet121, ResNet-50, and EfficientNetB0. Instead of training a deep classifier end-to-end, features were extracted and combined, and classification was performed using classical models such as Random Forest and Support Vector Machines.

This approach was particularly effective given the limited dataset size (~446 images), where fully trained deep networks tend to overfit or become unstable. The hybrid model achieved an accuracy of approximately 82.22% with AUC around 0.75–0.76. Performance was comparable to the best ResNet model while offering improved stability. However, increasing complexity through ensembles did not lead to further improvements.

## 4. Results and Analysis

The progression of models throughout this project demonstrates a clear and structured improvement in both accuracy and AUC, reflecting a deeper understanding of the problem and the modelling process.

The initial models exhibited low performance, with accuracy around 34% and AUC close to 0.50, indicating that the models were unable to effectively distinguish between classes. As training strategies were refined and class imbalance was addressed, gradual improvements were observed, with accuracy increasing to the range of 40–60% and AUC improving to approximately 0.60–0.65.

A significant turning point occurred when preprocessing was corrected to align with the expectations of pretrained networks. This adjustment led to a sharp increase in performance, with accuracy reaching approximately 83% and AUC approaching 0.80. This demonstrates that proper data handling and feature alignment can have a greater impact than architectural changes alone.

Further experimentation with alternative architectures such as DenseNet and hybrid approaches confirmed that the models had reached a performance plateau. These results reinforce the observation that once a strong feature representation is achieved, additional complexity yields diminishing returns.

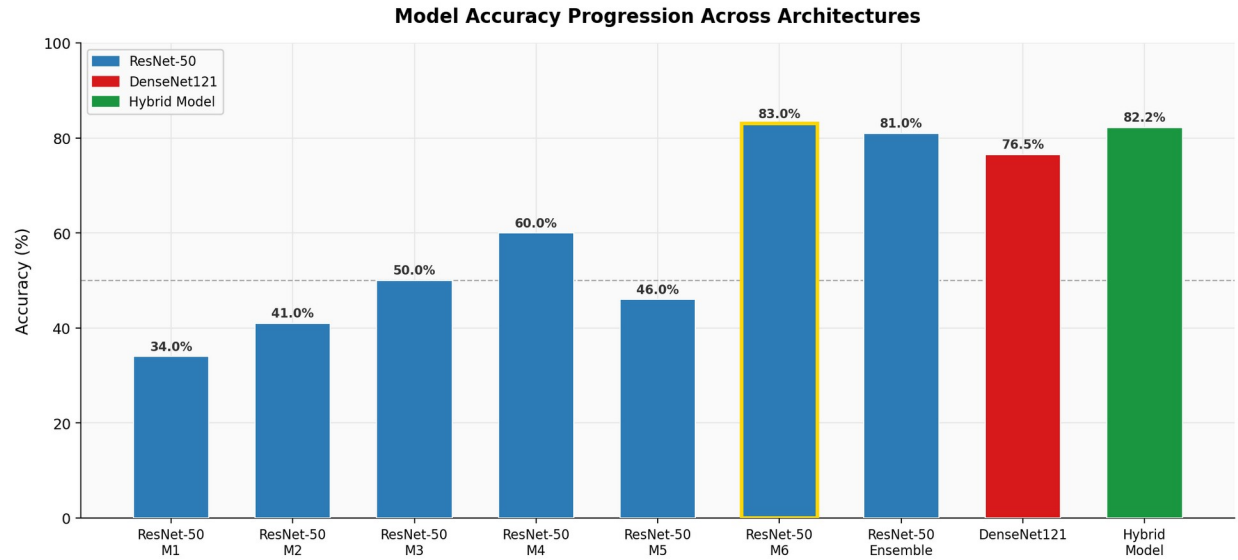


Figure 1: Accuracy (%) across all model iterations grouped by architecture

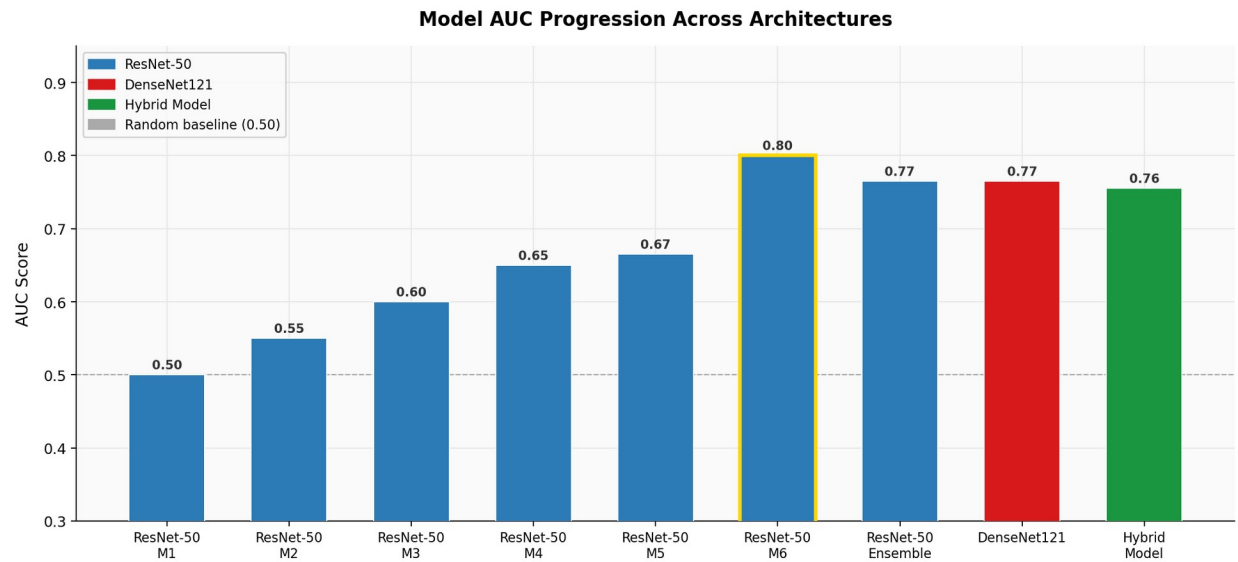


Figure 2: AUC score across all model iterations grouped by architecture

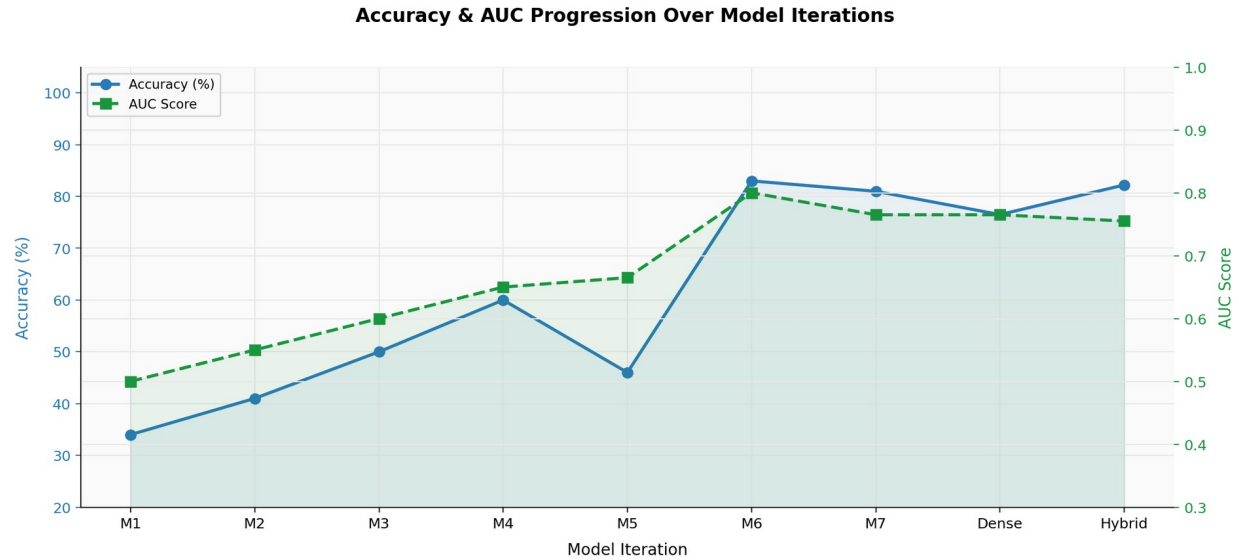


Figure 3: Combined Accuracy & AUC progression over model iterations

Overall, the progression highlights a systematic improvement driven by iterative refinement rather than trial-and-error experimentation. Each stage contributed to identifying key factors influencing performance, ultimately leading to a robust and well-understood model.

## 5. Conclusion

This project demonstrates the importance of systematic experimentation and refinement in developing effective machine learning models. Starting from initial implementations with limited performance, successive improvements in preprocessing, training strategy, and feature learning led to substantial gains in both accuracy and AUC.

The best results were achieved using a properly configured ResNet-50 model, which consistently outperformed more complex alternatives. While DenseNet-based models and hybrid approaches provided competitive performance, they did not significantly exceed the optimised ResNet pipeline.

An important insight from this work is that increasing model complexity does not necessarily lead to better results, particularly when working with limited data. Instead, careful preprocessing, appropriate use of transfer learning, and stable training methodologies play a more critical role in achieving high performance.

Additionally, the use of an auxiliary ageing dataset for pretraining highlights the value of leveraging related data to improve feature representation in low-data scenarios.

In conclusion, this work successfully demonstrates that deep learning-based approaches can be effectively applied to malnutrition detection using facial images, and that well-optimised models can achieve strong performance even under dataset constraints. The findings provide a foundation for further exploration and potential real-world deployment of such systems.